No singularity phase transition for Haagerup functions on free groups

Full counterexample packet for arXiv:1705.05457

11 August 2026

Status and answer. **Both advertised questions have negative answers.** Let $k \geq 2$, let $\mathbb{F}_k$ have its standard word length, and put $f_r(x) = r^{|x|}$ for $0 < r < 1$. Then

$$\delta_e \ll f_r.$$

Consequently f_r is not singular to $\mathrm{A}(\mathbb{F}_k)$ and is not singular to $\mathrm{B}_\lambda(\mathbb{F}_k)$ for *any* $0 < r < 1$. In particular this holds at $r = (2k-1)^{-1/2}$ and throughout the supercritical interval. Thus the suggested singularity phase transition does not occur.

The source questions

The final item of arXiv:1705.05457 (source PDF page 42) asks whether the critical Haagerup function is singular to the Fourier algebra and whether f_r is singular to the reduced Fourier–Stieltjes algebra above the critical value. The exact source paragraph is reproduced here from the archived PDF:

> (10) There is still not much known about properties of the Haagerup functions f_r discussed at the beginning of Section 6. There are two specific problems that we would like to advertise: Is $f_{\frac{1}{\sqrt{2k-1}}}$ singular to $\mathrm{A}(\mathbb{F}_k)$? Is f_r singular to $B_\lambda(\mathbb{F}_k)$ for all $r > \frac{1}{\sqrt{2k-1}}$? If it is so, it would provide a nice example of a 'phase transition' at point $r = \frac{1}{\sqrt{2k-1}}$ – below it the functions belong to the Fourier algebra and above that level they are singular even to the reduced Fourier–Stieltjes algebra.

The source defines $\varphi \ll \psi$ by $\mathrm{zs}(\varphi) \leq \mathrm{zs}(\psi)$ and $\varphi \perp \psi$ by $\mathrm{zs}(\varphi)\,\mathrm{zs}(\psi) = 0$ (Definition 4.7). For a discrete group, it calls f singular when $f \perp \delta_e$ (Definition 4.20); $\mathrm{A}(G)$ is the closed translation-invariant subspace generated by δ_e. Proposition 4.10 says that a functional is absolutely continuous with respect to another exactly when it is a norm limit of linear combinations of its two-sided translates.

The translate identity

Let S be the standard symmetric generating set of $\mathbb{F}_k$, so $|S| = 2k$, and write $q = 2k-1$. For left translation $(L_s f)(x) = f(s^{-1}x)$, the following identity holds pointwise on $\mathbb{F}_k$:

$$\sum_{s\in S} L_s f_r = (r^{-1} + qr) f_r - \frac{1-r^2}{r}\delta_e. \tag{1}$$

Equivalently,

$$\delta_e = \frac{r}{1-r^2}\left((r^{-1} + qr) f_r - \sum_{s\in S} L_s f_r\right). \tag{2}$$

Theorem. *For every $k \geq 2$ and $0 < r < 1$, one has $\delta_e \ll f_r$. Therefore $f_r \not\perp \mathrm{A}(\mathbb{F}_k)$ and $f_r \not\perp \mathrm{B}_\lambda(\mathbb{F}_k)$.*

Proof intuition. Away from the identity, multiplying a reduced word on the left by the $2k$ generators has a rigid pattern: one choice cancels its first letter, while the other $2k-1$ choices add a letter. Thus the adjacency sum of the radial function $r^{|x|}$ is an eigenfunction relation everywhere except at the root of the Cayley tree. Its defect at the root is a nonzero multiple of δ_e. Solving for that defect puts δ_e in the finite translate span of f_r, which is exactly what the source's absolute-continuity criterion requires.

Proof. Fix $x \neq e$ with $|x| = n$. Exactly one $s \in S$ cancels the first letter of x in $s^{-1}x$, giving length $n-1$; the other q choices give length $n+1$. Hence

$$\sum_{s\in S} f_r(s^{-1}x) = r^{n-1} + qr^{n+1} = (r^{-1} + qr)f_r(x).$$

At $x = e$, the left side is $(q+1)r$, whereas $(r^{-1}+qr)f_r(e) = r^{-1}+qr$. Their difference is $r-r^{-1} = -(1-r^2)/r$, proving (1) and (2).

Equation (2) expresses δ_e as a *finite* linear combination of f_r and its left translates. Proposition 4.10 of the source therefore gives $\delta_e \ll f_r$, i.e. $0 \neq \mathrm{zs}(\delta_e) \leq \mathrm{zs}(f_r)$. In particular $\mathrm{zs}(f_r)\,\mathrm{zs}(\delta_e) = \mathrm{zs}(\delta_e) \neq 0$, so $f_r \not\perp \delta_e$ and hence is not singular to $\mathrm{A}(\mathbb{F}_k)$. Finally $\delta_e \in \mathrm{A}(\mathbb{F}_k) \subseteq \mathrm{B}_\lambda(\mathbb{F}_k)$, so the same nonzero central-support overlap rules out singularity to $\mathrm{B}_\lambda(\mathbb{F}_k)$. □

Consequences and scope

The source records three membership regimes: $f_r \in \mathrm{A}(\mathbb{F}_k)$ below the critical value, $f_r \in \mathrm{B}_\lambda(\mathbb{F}_k)$ at the critical value, and $f_r \notin \mathrm{B}_\lambda(\mathbb{F}_k)$ above it. The theorem shows that this membership threshold is *not* a singularity threshold. Above the threshold, f_r is outside $\mathrm{B}_\lambda(\mathbb{F}_k)$ but still has nonzero central-support overlap with its element δ_e.

No approximation or limiting argument is involved, and the result covers the full parameter interval in which the Haagerup functions in the source are defined. The identity also works for $k = 1$, but that abelian case is outside the advertised free-group questions.

Verification. The word-counting proof was independently audited at the identity and at nonidentity words. An exact-rational checker exhaustively tests all reduced words of length at most six for nine pairs (k, r). Source PDF pages 18–19, 24, 33–34, and 42 were checked for the definitions, translate criterion, membership facts, and exact questions.

Novelty and human review. Searches by the exact question, by the adjacency identity, and through later work on radial positive-definite functions found no published resolution as of 11 August 2026. This is therefore recorded as a new full counterexample, subject to expert review. A reviewer should check the translation convention in Proposition 4.10, the root coefficient in (1), and the interpretation of singularity for a subspace.

References

[1] P. Ohrysko and M. Wasilewski, *Spectral theory of Fourier–Stieltjes algebras*, arXiv:1705.05457v4. See Definitions 4.7 and 4.20, Proposition 4.10, and final question (10).